TITLE PAGE

Smart system Badge

Student: Erasyl Bisembaev

Group: cs 24-16

Course: Introduction to IoT

Date: February 11, 2026

1. Introduction

1.1 Project Description

The **"SMART SYSTEM BADGE"** is an advanced IoT prototype designed for automated area monitoring and emergency containment. Unlike basic proximity sensors, this system combines distance measurement (Ultrasonic), human motion detection (PIR), and physical security (Servo-lock) to create a multi-layered safety protocol.

1.2 Objective and Relevance

The objective is to eliminate human error in high-risk zones. The system ensures that if an unauthorized person enters a restricted "Safety Bubble" (150 cm), the system not only alerts the operator but also physically blocks access (Servo) and triggers a high-intensity alarm.

2. IoT Technology Overview

2.1 IoT Architecture

- **Perception Layer:** HC-SR04 (Distance), PIR Sensor (Motion), and Panic Button (Manual override).
- **Processing Layer (Edge):** Arduino Uno R3 calculates sensor fusion data (Distance + Motion) to prevent false positives.
- **Application Layer:** LCD 16x2 UI for status reporting and physical actuators (Servo, Motor, Buzzer) for immediate response.

3. Hardware Implementation

3.1 Components (Final Configuration)

1. **Microcontroller:** Arduino Uno R3.
2. **Sensors:** HC-SR04 (Ultrasonic), PIR Sensor (Motion).
3. **Visual Output:** LCD 16x2, Green LED, Red LED.
4. **Audible Output:** Piezo Buzzer (using `tone()` frequency).
5. **Actuators:** DC Motor (Vibration), Servo Motor (Safety Gate/Lock).
6. **Control:** Slide Switch (Armed/Disarmed Mode), Push Button (Panic Alarm).
7. **Protection:** Resistors (220Ω for LEDs).

3.2 Hardware Pin Configuration (Detailed)

This configuration utilizes a strategic mix of Digital, Analog, and Pulse Width Modulation (PWM) pins to ensure the stable operation of all 10 system components while avoiding signal interference.

Component	Pin	Mode	Signal Type	Technical Function
LCD RS	D12	Output	Digital	Register Select: Toggles between command and data mode.
LCD Enable	D6	Output	Digital	Starts the data read/write sequence for the display.
LCD D4-D7	5,4,3,2	Output	Digital	4-bit Parallel Bus: Sends character data to the screen.
Ultrasonic Trig	D9	Output	Digital	Trigger: Sends a 10μs sonic burst to start measurement.
Ultrasonic Echo	D10	Input	Digital	Echo: Measures "Time of Flight" (the duration of the high pulse).
Piezo Buzzer	D11	Output	PWM (Tone)	Frequency Control: Generates a 2500Hz square wave for the siren.
DC Motor	D7	Output	Digital	Actuator: Drives the vibration motor for haptic feedback.
Servo Motor	A2	Output	PWM	Mechanical Lock: Controls gate position (0° to 90°).
PIR Sensor	D8	Input	Digital	Perception: Detects IR signatures from human movement.
Red LED	D13	Output	Digital	Warning: Visual indicator for "Danger" or "Panic" states.
Green LED	A0	Output	Digital	Status: Visual indicator for "Safe" or "Armed" states.
Slide Switch	A1	Input	Pull-up	Mode Toggle: Determines if the security logic is active.
Panic Button	A3	Input	Pull-up	Manual Override: Force-triggers the alarm when grounded.

4. Methods and Algorithm

4.1 Mathematical Signal Processing

The system calculates the distance based on the physics of sound wave propagation. The HC-SR04 sensor emits an ultrasonic pulse that bounces off an object and returns to the receiver.

1. The microcontroller triggers the **Trig** pin.
2. The sensor sends 8 bursts of 40kHz ultrasound.
3. The **Echo** pin stays HIGH for the duration of the sound's travel.
4. The distance (S) is calculated using the constant speed of sound (343 m/s or 0.0343 cm /us):

$$S = \frac{\text{Duration} \times 0.0343}{2}$$

4.2 Algorithm Steps

- **ALARM STATE:**
  1. `tone(11, 2500)` is activated for maximum audible penetration.
  2. `digitalWrite(7, HIGH)` activates the haptic DC motor.
  3. `myServo.write(0)` rotates the lock to the 0° (Closed) position.
  4. LCD is updated with the message `!!! DANGER !!!`.
- **SAFE STATE:**
  1. `noTone(11)` silences the buzzer.
  2. Motor and Red LED are deactivated.
  3. `myServo.write(90)` rotates the lock to the 90° (Open) position.
  4. LCD displays current distance and PIR status for diagnostic monitoring.

Stage 5: Stabilization

- A 200ms software delay is implemented at the end of the loop to prevent "screen flickering" and to allow the power rail to stabilize between actuator movements, preventing the Arduino from resetting due to current spikes.

5. Testing and Results

5.1 System Stability

- **Issue:** Power instability when starting the DC Motor and Servo simultaneously.
- **Solution:** Implemented a common ground (GND) rail and added a `delay(200)` in the loop to stabilize the Arduino's internal voltage regulator.
- **Switch Fix:** Moved the Slide Switch from the power line to a Digital Input (A1) with `INPUT_PULLUP`. This prevents the LCD from crashing during mode switching.

6. Conclusion

The prototype successfully integrates 10+ components into a reliable industrial safety solution. By using sensor fusion (PIR + Ultrasonic), the system achieves higher accuracy than standard proximity badges.

Future Prospects: Integration of an NPN transistor for high-voltage industrial sirens and an ESP8266 for remote cloud logging.